

Visual Policy Learning for Panda Reach

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Technical report · MuJoCo Franka Panda reach · imitation + RL

Abstract

We study visuomotor reaching on a shared MuJoCo Franka Panda task: move the end-effector within 5 cm of a randomly sampled 3D goal under a dense distance-based reward. Visual imitation trains frozen-encoder behavior cloning (ResNet-18, CLIP, DINOv2-S) from 100 IK expert demos on 64×64 RGB; online RL trains PPO and SAC on privileged state with the same reward and actions, matched to Stable-Baselines3 under identical budgets. Closed-loop success is 100% / 80% / 60% for CLIP / ResNet / DINOv2-S (5 ep) and 100% / 96% for SAC (50k) / PPO (150k) (25 ep); SB3 is comparable. Lower BC validation loss does not track success; DINOv2-S is resolution-starved at 64×64; RGB policies are brittle under lighting and noise. Results separate vision failure from control and motivate richer observations and larger evals.

1. Headline results

Method	Learning	Obs	Success	Notes
CLIP BC	Imitation	RGB	100%	5 ep
ResNet BC	Imitation	RGB	80%	5 ep
DINOv2-S BC	Imitation	RGB	60%	5 ep
SAC	RL	state	100%	25 ep, 50k
PPO	RL	state	96%	25 ep, 150k
SB3 SAC	RL	state	96%	25 ep, 50k
SB3 PPO	RL	state	100%	25 ep, 150k
IK expert	Scripted	state	100%	~45 steps

RGB BC = vision+control; state RL = control alone. Modalities differ on purpose.

Closed-loop success comparison

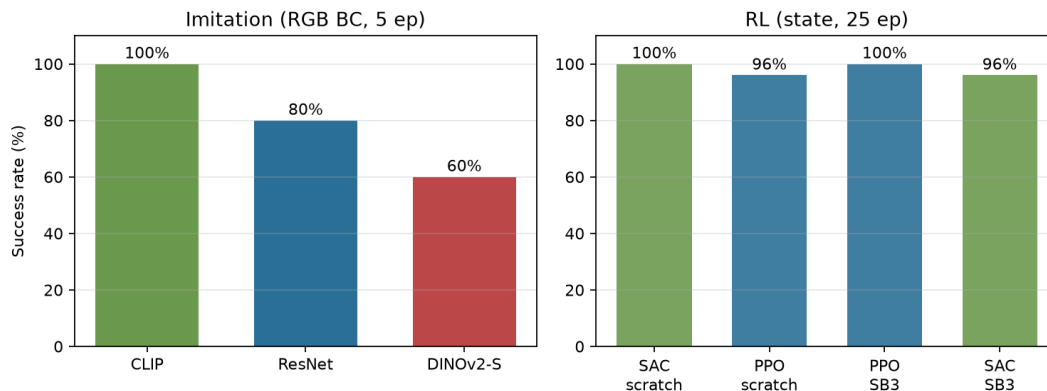


Fig. 1 — Closed-loop success: imitation (RGB, 5 ep) vs RL (state, 25 ep).

2. Imitation learning (frozen-encoder BC)

100 expert episodes → 1,541 transitions (stride=3). Train: 200 epochs, AdamW + cosine. Closed-loop eval: 5 episodes, 64×64 RGB.

Model	Feat dim	Best val loss	Success	Avg reward	Avg steps
CLIP	512	0.0234	100%	-10.59	43.0
ResNet-18	512	0.0177	80%	-14.72	56.2
DINOv2-S	384	0.0200	60%	-20.12	69.8

CLIP leads closed-loop success despite higher val loss. DINOv2-S is resolution-starved at 64×64.

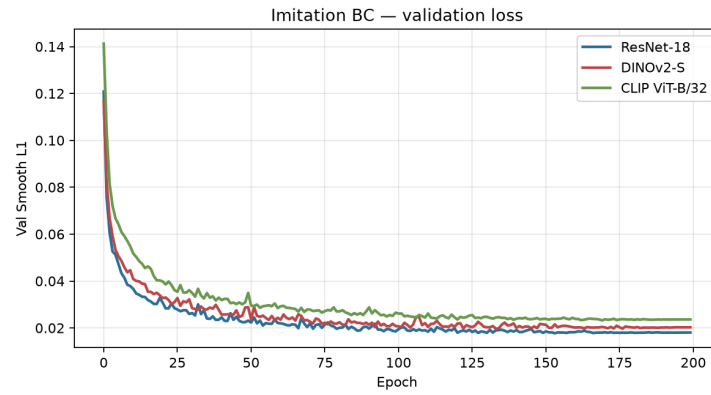


Fig. 2 — BC validation Smooth L1 loss (frozen encoders).

3. Reinforcement learning (privileged state)

PandaReachStateEnv. Shared task/reward/actions; budgets PPO 150k, SAC 50k. Eval: 25 ep, seed 42.

Method	Timesteps	Success	Avg reward	Avg steps
SAC	50k	100%	-9.23	42.6
PPO	150k	96%	-13.07	50.0
SB3 PPO	150k	100%	-8.57	35.9
SB3 SAC	50k	96%	-11.58	42.7

Ours = SB3 success. SAC is more sample-efficient; PPO uses obs/reward running normalization.

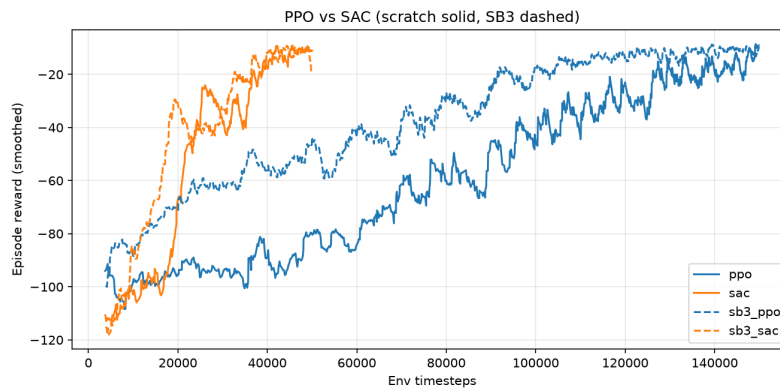


Fig. 3 — Episode reward vs timesteps (solid = ours, dashed = SB3).

4. Robustness (RGB perturbations)

Wrappers: dark ($\times 0.5$), bright ($\times 1.5$), occlusion (16×16), noise ($\sigma=15$). 1 episode/cell (high variance). Avg reward (higher better).

Condition	ResNet	DINOv2-S	CLIP
normal	-16.4	-9.2	-31.7
dark	-32.9	-30.9	-29.2
bright	-5.6	-38.3	-32.3
occlusion	-33.1	-6.1	-8.8
noise	-34.3	-42.7	-33.5

Lighting/noise brittle for all three.

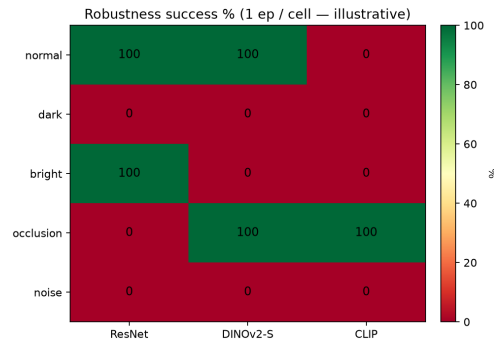


Fig. 4 — Robustness heatmap (1 ep/cell).

5. Takeaways, limitations, next steps

- CLIP strongest RGB BC at 64×64 ; lower val loss $\neq$ higher closed-loop success.
- Privileged-state PPO/SAC solvable; SAC sample-efficient; SB3 matches under same budgets.
- Limitations: $n=5$ IL eval; single RL seed; 1-ep robustness; frozen 64×64 (DINOv2 starved).
- Next: 15–25 ep IL; multi-seed RL $\pm$ std; proprio+RGB; DAgger; visual RL / light unfreeze.

Artifacts: README.md · report.pdf · results/metrics/ · videos/ · notebooks/